TMEM94-Associated Congenital Heart Defect–Facial Dysmorphism–Developmental Delay Syndrome (IDDCDF)

Disease: Intellectual Developmental Disorder with Cardiac defects and Dysmorphic Facies (IDDCDF) **MONDO:** MONDO:0032672 · **OMIM:** 618316 · **Causal gene: TMEM94** (17q25.1; HGNC:25151; UniProt Q12767; RefSeq NM_014738.6) **Category:** Mendelian, autosomal recessive · **Inheritance:** Biallelic truncating loss-of-function

Summary

IDDCDF is an ultra-rare autosomal recessive multisystem syndrome caused by biallelic truncating (loss-of-function) variants in the gene TMEM94. It is defined by a core clinical triad: (1) global developmental delay / intellectual disability, (2) congenital heart defects (present in ~90% of reported cases), and (3) a distinctive facial gestalt (triangular face, deep-set eyes, broad nasal root/tip, anteverted nostrils, thick arched eyebrows, hypertrichosis, pointed chin, and hypertelorism). The disorder was first delineated in 2018 by Stephen et al. in 10 affected individuals from 6 unrelated families of diverse ethnic backgrounds, and its gene–disease relationship has since been independently confirmed and extended to the prenatal/fetal phenotype and the overgrowth–intellectual-disability spectrum.

The molecular identity of the TMEM94 protein remained obscure until 2024, when Vishnu et al. demonstrated that **TMEM94 is "ERMA," an endoplasmic-reticulum–localized P-type ATPase that pumps magnesium (Mg^{2+}) into the ER lumen.** This discovery provides the strongest available mechanistic explanation for the cardiac phenotype: loss of ERMA impairs ER Mg^{2+} refilling, disrupting intracellular Mg^{2+}/Ca^{2+} homeostasis and producing abnormal cardiomyocyte Ca^{2+} cycling and cardiac dysfunction. In parallel, patient cells show reduced TMEM94 expression and broad transcriptional dysregulation of genes governing cell growth, proliferation, and survival — a plausible route to the neurodevelopmental and craniofacial features. The complete CRISPR/Cas9 knockout mouse is embryonic lethal, underscoring an essential developmental role.

A notable unresolved tension — a **dosage-sensitivity paradox** — emerges from the combined data. In mice, ERMA *haploinsufficiency* (Erma$^{+/-}$) already causes cardiac dysfunction and homozygous nulls are embryonic lethal; yet in humans the disease is strictly recessive, with heterozygous carriers unaffected and gnomAD showing tolerance of heterozygous loss-of-function (pLI≈0, LOEUF=0.61). Management of IDDCDF is entirely supportive (cardiac surgical repair, developmental/rehabilitative therapies) with genetic counseling and prenatal/preimplantation options for prevention; no disease-specific or molecularly targeted therapy currently exists, although the ERMA/Mg^{2+} mechanism nominates magnesium homeostasis as a hypothesis-generating therapeutic avenue.

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1. Disease Information

Overview. IDDCDF (Intellectual Developmental Disorder with Cardiac defects and Dysmorphic Facies) is a Mendelian, autosomal recessive neurodevelopmental syndrome with prominent congenital cardiac involvement and a recognizable facial gestalt. It is caused by biallelic truncating variants in **TMEM94**, which encodes a highly conserved multi-pass transmembrane protein now known to be the endoplasmic-reticulum magnesium transporter ERMA.

Key identifiers.

Resource	Identifier
OMIM (disease)	618316
MONDO	MONDO:0032672
Gene (HGNC)	TMEM94 / HGNC:25151
Gene locus	17q25.1
UniProt	Q12767
RefSeq transcript	NM_014738.6
InterPro (protein domain)	IPR023298 (P-type ATPase, transmembrane-domain superfamily)
ICD-10 / ICD-11	No specific code; subsumed under general congenital-malformation / intellectual-disability categories
MeSH	No specific descriptor
Orphanet	No dedicated ORPHAcode identified for the TMEM94-specific entity

Synonyms / alternative names. IDDCDF; TMEM94-associated neurodevelopmental disorder; TMEM94-related congenital heart defect–facial dysmorphism–developmental delay syndrome.

Information source. Disease-level knowledge is derived from aggregated case reports and small cohort studies (individual-patient clinical and molecular data), not from large EHR-based or registry datasets. The evidence base is a small number of primary publications describing a total of roughly a dozen patients worldwide.

2. Etiology

Primary cause (genetic). IDDCDF is a monogenic disorder caused by **biallelic truncating loss-of-function (LoF) variants in TMEM94**. Both homozygous and compound-heterozygous genotypes have been reported. The founding study identified such variants in 10 affected individuals from 6 unrelated families "of different ethnic origins," with segregation consistent with autosomal recessive inheritance (PMID: 30526868).

> *"we describe ten similarly affected individuals from six unrelated families of different ethnic origins having bi-allelic truncating variants in TMEM94, which encodes for an uncharacterized transmembrane nuclear protein that is highly conserved across mammals"* (PMID: 30526868).

Genetic risk factors. The only established risk factor is possession of two pathogenic TMEM94 LoF alleles. **Consanguinity** increases the risk of homozygosity for rare recessive alleles and is relevant here, as several original families were consanguineous. There are no known common susceptibility loci or GWAS associations (not applicable to a rare Mendelian disorder).

Modifier genes. None established. Phenotypic variability (e.g., the single individual without CHD in the original cohort) is documented, but no modifier locus has been mapped.

Environmental risk / protective factors and gene–environment interactions. None identified. As a genotype-determined recessive Mendelian disorder, there are no established environmental risk or protective factors and no documented gene–environment interactions (not applicable / no data).

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3. Phenotypes

The phenotype centers on a **triad**: developmental/cognitive impairment, congenital heart defects, and distinctive craniofacial dysmorphism, with additional multisystem features. Frequencies below derive from the original cohort (Stephen et al. 2018, n=10) (PMID: 30526868), with supporting observations from Al-Hamed et al. 2020 (PMID: 32825426) and Yüksel Ülker et al. 2023 (PMID: 36919607).

Phenotype	Type	Frequency	Onset	Suggested HPO term
Global developmental delay / intellectual disability	Behavioral/ cognitive	All (10/10)	Neonatal–childhood	HP:0001263 / HP:0001249
Congenital heart defect	Physical/ structural	9/10 (~90%)	Congenital	HP:0001627
Triangular face	Craniofacial	Frequent	Congenital	HP:0000325
Deep-set eyes	Craniofacial	Frequent	Congenital	HP:0000490
Broad nasal root/tip	Craniofacial	Frequent	Congenital	HP:0000431 / HP:0000455
Anteverted nares	Craniofacial	Frequent	Congenital	HP:0000463
Thick, arched eyebrows	Craniofacial	Frequent	Congenital	HP:0000574 / HP:0002553
Hypertrichosis	Integumentary	Frequent	Congenital	HP:0000998
Pointed chin	Craniofacial	Frequent	Congenital	HP:0000307
Hypertelorism	Craniofacial	Frequent	Congenital	HP:0000316
Birthweight in upper-normal range / overgrowth tendency	Growth	Most	Prenatal/ birth	HP:0001520

> *"global developmental delay/intellectual disability, and dysmorphic facial features including triangular face, deep set eyes, broad nasal root and tip and anteverted nostrils, thick arched eye brows, hypertrichosis, pointed chin, and hypertelorism. Birthweight in the upper normal range was observed in most, and all but one had congenital heart defects (CHD)"* (PMID: 30526868).

Severity, progression, and QoL. The disorder is congenital and static in character (a malformation/neurodevelopmental syndrome rather than a progressive degenerative one). Severity is variable but generally moderate-to-severe given universal developmental delay plus structural heart disease. Quality-of-life impact is substantial — cognitive impairment affects

education, communication, and independent living, while congenital heart disease carries surgical and functional morbidity. Formal QoL instruments (EQ-5D, SF-36, PROMIS) have not been applied to this ultra-rare disorder (no data).

Phenotype expansion. Al-Hamed et al. (2020) extended the spectrum to **prenatal/fetal anomalies** detectable on antenatal ultrasound, and Yüksel Ülker et al. (2023) detected a TMEM94 pathogenic variant in an **overgrowth–intellectual-disability (OGID)** cohort, nudging the phenotype toward the overgrowth/growth-pathway syndromes.

> *"expanded the phenotypes of very rare OGID syndromes caused by FIBP and TMEM94"* (PMID: 36919607).

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4. Genetic / Molecular Information

Causal gene. TMEM94 (17q25.1; HGNC:25151; UniProt Q12767; reference transcript NM_014738.6). The encoded protein is 1,356 amino acids, a multi-pass (10 transmembrane segment) ER protein of the P-type ATPase superfamily (see Section 6).

Pathogenic variants — all truncating LoF. All reported pathogenic/likely-pathogenic variants are truncating (nonsense, frameshift, or canonical splice-site); **no pathogenic missense variant has been reported.** ClinVar (NM_014738.6) lists 8 pathogenic/likely-pathogenic variants:

HGVS (coding)	Protein consequence	Variant class	Classification
c.3776G>A	p.Trp1259Ter	Nonsense	Pathogenic
c.1626C>G	p.Tyr542Ter	Nonsense	Likely pathogenic
c.657del	p.Phe220fs	Frameshift	Likely pathogenic
c.2129dup	p.Leu710fs	Frameshift	Likely pathogenic
c.2280_2281del	p.Gln761fs	Frameshift	Likely pathogenic
c.33_42del	p.Pro12fs	Frameshift	Likely pathogenic
c.410-2A>T	Splice acceptor	Splice-site	Likely pathogenic
c.273-1G>C	Splice acceptor	Splice-site	Likely pathogenic

Among 404 ClinVar submissions for TMEM94, the majority are benign/likely-benign or VUS, consistent with a **loss-of-function-only** pathogenicity model in a recessive gene (F006).

Functional consequence. Loss of function via nonsense-mediated decay (NMD) of truncating transcripts. Patient cells show **reduced TMEM94 expression** — *"Gene expression analysis in available cells from affected individuals showed reduced expression of TMEM94"* (PMID: 30526868). This is a classic recessive LoF mechanism; both alleles must be disrupted to cause disease.

Allele frequency / constraint. gnomAD shows TMEM94 to be tolerant of heterozygous LoF (pLI≈0; LOEUF/oe_lof_upper = 0.61; observed/expected LoF = 81/158.6), consistent with unaffected heterozygous carriers. Individual pathogenic alleles are ultra-rare or absent in population databases.

Origin. Germline. No somatic involvement reported (not applicable — not a cancer gene in this context).

Modifier genes / epigenetics / chromosomal abnormalities. No modifier loci, disease-specific epigenetic signatures, or recurrent chromosomal rearrangements described (no data). Chromosomal microarray was used diagnostically in the prenatal setting primarily to exclude CNVs.

5. Environmental Information

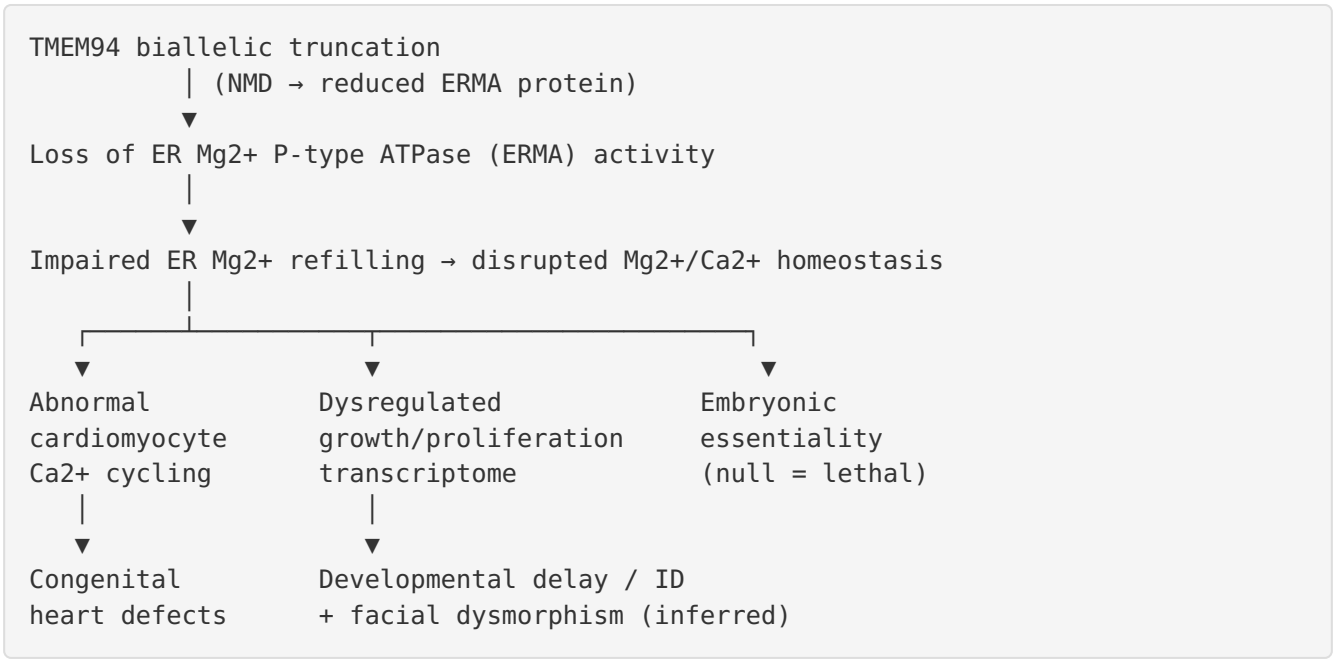
No environmental, lifestyle, toxicological, or infectious contributors are known or expected. IDDCDF is a purely genetic Mendelian disorder; environmental factors (toxins, radiation, diet, smoking, infectious agents) are **not applicable**. Consanguinity is a population-genetic, not environmental, risk amplifier.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic truncating variant in TMEM94** introduces a premature stop / frameshift / splice defect → **leads to** nonsense-mediated decay and **reduced/absent TMEM94 (ERMA) protein** (*demonstrated*: reduced patient-cell expression).
2. Loss of ERMA, an ER-localized P-type ATPase bearing a GMN motif, **results in** failure of **Mg^{2+} transport into the endoplasmic reticulum** (*demonstrated* biochemically for ERMA; *inferred* to apply to patient truncating alleles).
3. Impaired ER Mg^{2+} refilling **leads to** disrupted intracellular **Mg^{2+} homeostasis**, which **results in** abnormal **Ca^{2+} cycling** (Mg^{2+} and Ca^{2+} handling are coupled at the ER) (*demonstrated* in $Erma^{+/-}$ cardiomyocytes and ERMA-silenced iPSC-cardiomyocytes).
4. **Branch A — Cardiac:** Abnormal cardiomyocyte Ca^{2+} cycling **leads to** contractile/electrical dysfunction and, developmentally, to **congenital heart defects** and cardiac dysfunction (*strongest mechanistic link*; demonstrated at cellular level, inferred for the structural CHD).
5. **Branch B — Growth/neurodevelopment:** Reduced ERMA **results in** broad transcriptional **dysregulation of cell-growth, proliferation, and survival genes** (*demonstrated* by microarray + RNA-seq), **predicted to impact** neurodevelopmental and hematological systems → contributing to **developmental delay / intellectual disability** and, plausibly, **craniofacial dysmorphism** (*inferred*).
6. **Branch C — Essentiality:** Complete loss (homozygous null in mouse) **leads to embryonic lethality**, establishing ERMA as developmentally essential; human viable states imply retained partial function (*demonstrated* in mouse; human residual function *inferred*).



Supporting detail by category

- **Molecular pathways / protein function.** TMEM94 = **ERMA**, an ER multi-pass transmembrane P-type ATPase (UniProt Q12767; 1,356 aa; 10 TM segments; InterPro IPR023298). It possesses the canonical large cytosolic **actuator (A), nucleotide-binding (N), and phosphorylation (P) domains** of P-type ATPases and *"uniquely combines a P-type ATPase domain and a GMN motif for ER Mg^{2+} uptake"* (PMID: 38513662). A conserved tyrosine is crucial for Mg^{2+} binding/activity; the mechanism is conserved from prokaryotic mgtA/mgtB.
- **Cellular processes.** Ion transport (Mg^{2+}) across the ER membrane; ER ion homeostasis; coupled Ca^{2+} cycling; cell growth/proliferation/survival signaling (via the dysregulated transcriptome).
- **Metabolic / biochemical.** Mg^{2+} is a cofactor for hundreds of ATP-dependent enzymes; ER Mg^{2+} depletion plausibly perturbs ATPase/kinase activity and protein folding, though the downstream metabolic details in IDDCDF are not directly characterized (*inferred*).
- **Tissue damage.** Cardiomyocyte dysfunction via Ca^{2+} mishandling is the demonstrated injury mechanism; neuronal/craniofacial consequences are inferred from the transcriptomic data.
- **Molecular profiling.** *"Global transcriptome profiling using microarray and RNA sequencing revealed several dysregulated genes essential for cell growth, proliferation and survival that*

are predicted to have an impact on cardiotoxicity hematological system and neurodevelopment" (PMID: 30526868).

- **Functional validation.** *"Cardiac dysfunction by haploinsufficiency, abnormal Ca^{2+} cycling in mouse $Erma^{+/-}$ cardiomyocytes, and ERMA mRNA silencing in human iPSC-cardiomyocytes collectively define ERMA as an essential component of ER Mg^{2+} uptake in eukaryotes"* (PMID: 38513662).

Suggested ontology terms. GO: magnesium ion transmembrane transport (GO:1903830), calcium ion transport (GO:0006816), regulation of cardiac muscle contraction (GO:0055117), heart development (GO:0007507), endoplasmic reticulum membrane (GO:0005789). CL: cardiac muscle cell / cardiomyocyte (CL:0000746), neuron (CL:0000540). CHEBI: magnesium(2+) (CHEBI:18420), calcium(2+) (CHEBI:29108).

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7. Anatomical Structures Affected

Organ level. - **Heart** (primary) — congenital structural defects and cardiac dysfunction. UBERON:0000948. - **Brain / central nervous system** (primary) — developmental delay and intellectual disability. UBERON:0000955. - **Craniofacial skeleton / face** (primary) — dysmorphism. UBERON:0001456 (face). - **Hematological system** — predicted secondary involvement from transcriptomic analysis.

Body systems: cardiovascular, nervous, and musculoskeletal/craniofacial, with possible hematologic involvement.

Tissue and cell level. Cardiac muscle tissue and **cardiomyocytes** (CL:0000746) are the best-documented affected cell type (demonstrated Ca^{2+}-cycling abnormalities). Neural tissue / **neurons** (CL:0000540) are implicated by the developmental phenotype (*inferred*).

Subcellular level. The **endoplasmic reticulum** (GO:0005783; ER membrane GO:0005789) is the key compartment, as ERMA resides in and functions at the ER membrane. The protein was originally described as a transmembrane nuclear-associated protein; the definitive, functionally validated localization is the ER.

Localization / lateralization. Cardiac defects and facial features are midline/bilateral congenital malformations; no consistent lateralization is reported.

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8. Temporal Development

- **Onset:** Congenital. Structural heart defects and facial dysmorphism are present at birth (and detectable prenatally per Al-Hamed et al. 2020); developmental delay becomes apparent in infancy/early childhood.
- **Onset pattern:** Congenital/static (malformation syndrome), not acute or relapsing.
- **Progression:** Non-progressive in the degenerative sense; structural and cognitive features are fixed, though developmental trajectories and cardiac status evolve with growth and surgical intervention.
- **Course / duration:** Chronic, lifelong.
- **Critical period:** Embryonic/fetal cardiac and craniofacial morphogenesis is the window of vulnerability; the embryonic lethality of the complete mouse knockout underscores the essential early-developmental role. There is no established therapeutic window to reverse structural defects once formed.

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9. Inheritance and Population

- **Inheritance pattern:** Autosomal recessive (biallelic truncating LoF). Heterozygous carriers are unaffected.
- **Penetrance / expressivity:** Appears high/complete for the core phenotype in biallelic individuals, with **variable expressivity** (e.g., 1/10 without CHD; variability in facial and growth features).
- **Epidemiology:** Ultra-rare. Prevalence and incidence are not formally quantified; the world literature comprises roughly a dozen reported individuals. No reliable per-100,000 estimate exists (no data).
- **Founder effects / geographic distribution:** No specific founder allele established; original families were of diverse ethnic origins, several consanguineous.
- **Consanguinity:** A relevant risk amplifier for homozygosity in affected families.
- **Carrier frequency:** Not formally established; gnomAD het-LoF tolerance (pLI≈0, LOEUF=0.61) indicates carriers exist in the general population without phenotype.
- **Sex ratio / age distribution:** No sex predilection reported (autosomal); affected individuals present from the prenatal period through childhood.

10. Diagnostics

Genetic testing is the definitive diagnostic modality. - **Whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)**, typically as **trio** analysis, is the primary route to diagnosis; it identified the gene in all published cohorts. Trio-ES combined with chromosomal microarray was used in the prenatal setting (PMID: 32825426).

> *"we used trio-exome sequencing (trio-ES) in conjunction with chromosomal microarray analysis (CMA) to identify two novel homozygous loss of function variants"* (PMID: 32825426).

- **Gene panels / single-gene testing:** TMEM94 is included on broad intellectual-disability and congenital-heart-defect NGS panels; targeted single-gene testing is appropriate for familial variant confirmation.
- **Chromosomal microarray (CMA):** Useful to exclude CNVs, especially prenatally, but does not detect the point/indel LoF variants causal here.
- **Variant interpretation:** Per ACMG/AMP, truncating variants in this gene are classified pathogenic/likely-pathogenic (ClinVar, NM_014738.6).

Clinical / imaging work-up. - **Echocardiography** (and fetal echocardiography/anomaly ultrasound prenatally) to define congenital heart defects. - **Developmental assessment** for delay / intellectual disability. - **Dysmorphology examination** for the characteristic facial gestalt. - Brain MRI, ECG, and other studies as clinically indicated.

Biomarkers / omics diagnostics. No validated circulating biomarker exists. Research-level transcriptomic profiling has been mechanistically informative but is not a clinical diagnostic. The ERMA/Mg^{2+} biology raises the untested possibility of Mg^{2+}-handling assays as future biomarkers (*speculative*).

Differential diagnosis. Other syndromic congenital-heart-defect-with-developmental-delay disorders (e.g., Kabuki syndrome, CHARGE syndrome, Noonan/RASopathies, Cornelia de Lange spectrum, and overgrowth–ID syndromes such as those caused by FIBP/SUZ12). The specific facial gestalt plus biallelic TMEM94 LoF genotype distinguishes IDDCDF.

Screening. Cascade carrier testing within affected families; prenatal/preimplantation testing where the familial variants are known. No population newborn-screening program exists for this ultra-rare disorder.

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11. Outcome / Prognosis

Formal survival, mortality, and natural-history data are **not available** for this ultra-rare disorder. Prognosis is driven principally by the **severity of the congenital heart defect** (and its surgical correctability) and the **degree of developmental/intellectual impairment**. The complete-null state is embryonic lethal in mice, implying that viable patients retain partial gene function. Morbidity is substantial and lifelong, encompassing cognitive disability and cardiac disease. No validated prognostic biomarkers exist. Recovery of the structural/cognitive deficits is not expected; management aims at functional optimization.

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12. Treatment

No disease-specific or molecularly targeted therapy exists. Management is **entirely supportive and symptom-directed**:

- **Cardiac:** Surgical/interventional repair of congenital heart defects; cardiology follow-up and heart-failure management as needed. (NCIT: Cardiac Surgical Procedure.)
- **Neurodevelopmental:** Early-intervention programs; physical, occupational, and speech therapy; special education. (NCIT: Physical Therapy; Occupational Therapy; Speech Therapy; Rehabilitation Therapy.)
- **Multidisciplinary care:** Genetics, cardiology, developmental pediatrics/neurology, and dysmorphology.
- **Genetic counseling:** For recurrence-risk assessment and family planning (25% recurrence risk for carrier couples). (NCIT: Genetic Counseling.)

Pharmacotherapy / advanced therapeutics. No approved pharmacologic, gene, cell, or RNA-based therapy. No pharmacogenomic guidance is established. The 2024 identification of ERMA as an ER Mg^{2+} transporter nominates **magnesium homeostasis** as a mechanistic hypothesis for future therapeutic exploration, but this is entirely untested and should be regarded as a research direction, not a recommendation.

Experimental treatments / clinical trials. None identified for IDDCDF (no NCT trials specific to TMEM94 disease).

13. Prevention

- **Primary prevention:** Not applicable to a genetic disorder beyond reproductive planning. **Genetic counseling** is the cornerstone; consanguinity counseling is relevant in at-risk populations.
- **Genetic/reproductive prevention: Carrier screening** of relatives, **prenatal diagnosis** (where familial variants are known), and **preimplantation genetic testing (PGT-M)** can prevent recurrence in known-carrier couples. Trio-ES plus CMA enabled prenatal detection (PMID: 32825426).
- **Secondary/tertiary prevention:** Early echocardiographic and developmental assessment to guide timely cardiac surgery and early-intervention therapies, minimizing complications and optimizing function.
- **Immunization / behavioral / public-health / environmental interventions:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** TMEM94/ERMA is **highly conserved across mammals**, and the Mg^{2+}-transport mechanism is conserved down to prokaryotes (bacterial mgtA/mgtB P-type ATPases) (PMID: 38513662). The mouse ortholog is *Tmem94/Erma* (NCBI Taxon: *Mus musculus*, 10090).
- **Natural disease in other species:** No naturally occurring companion-animal or wildlife disease attributable to TMEM94 has been catalogued (no OMIA entry identified). (No data.)
- **Comparative biology:** The deep evolutionary conservation of the P-type ATPase + GMN Mg^{2+}-uptake mechanism underscores the fundamental, conserved role of ER Mg^{2+} homeostasis.

15. Model Organisms

Model	Type	Key finding	Reference
CRISPR/Cas9 *Tmem94* knockout mouse (homozygous null)	Mammalian, germline KO	**Embryonic lethal** — establishes developmental essentiality	PMID: 30526868
Erma$^{+/-}$ heterozygous mouse	Mammalian, haploinsufficient	**Cardiac dysfunction**; abnormal cardiomyocyte Ca^{2+} cycling	PMID: 38513662
Human iPSC-derived cardiomyocytes, ERMA mRNA silencing	In vitro cellular	Recapitulates Ca^{2+}-cycling/ contractile defects	PMID: 38513662
Patient-derived cells (expression/ transcriptomics)	In vitro	Reduced TMEM94 expression; dysregulated growth/ proliferation transcriptome	PMID: 30526868

Phenotype recapitulation & limitations. The mouse and iPSC models robustly recapitulate the **cardiac** dimension of the disorder, directly linking ERMA loss to cardiomyocyte Ca^{2+} dysfunction. However, the **homozygous-null mouse is embryonic lethal**, so it does not model the viable human phenotype, and no mouse model reproduces the full triad including the neurodevelopmental and craniofacial features. This is the key limitation and the source of the dosage paradox below.

Mechanistic Model / Interpretation

The combined evidence supports a coherent model in which **IDDCDF is an ER-magnesium-homeostasis disorder.** Biallelic truncation of TMEM94 removes the ERMA P-type ATPase that pumps Mg^{2+} into the ER; the resulting failure to refill ER Mg^{2+} destabilizes coupled Mg^{2+}/Ca^{2+} handling. In cardiomyocytes — exquisitely dependent on tight Ca^{2+} cycling — this produces the demonstrated contractile/electrical dysfunction and underlies the congenital heart defects (the best-supported causal branch). In parallel, loss of ERMA broadly dysregulates genes governing cell growth, proliferation, and survival, a plausible (though not mechanistically dissected)

route to the neurodevelopmental delay and craniofacial dysmorphism. The embryonic lethality of the complete knockout confirms ERMA as developmentally essential, consistent with a congenital, multisystem syndrome.

The dosage-sensitivity paradox (F007)

Observation	Mouse	Human
Heterozygous LoF	Cardiac dysfunction ($Erma^{+/-}$)	**Unaffected carrier** (gnomAD het-LoF tolerant, pLI≈0, LOEUF=0.61)
Biallelic / null	Embryonic lethal	**Viable with IDDCDF** (recessive)

This cross-species discordance implies either (a) species differences in dosage sensitivity of ER Mg^{2+} transport, (b) residual/partial function from human truncating alleles (e.g., escape from NMD, truncated proteins retaining partial activity, or compensatory redundancy), or (c) differences in the mouse $Erma^{+/-}$ cardiac-assay sensitivity versus human clinical thresholds. Resolving it is the central open mechanistic question.

Evidence Base

Paper	PMID	Role / contribution
Bi-allelic TMEM94 Truncating Variants Are Associated with Neurodevelopmental Delay, Congenital Heart Defects, and Distinct Facial Dysmorphism (Stephen et al., 2018)	30526868	**Founding gene–disease paper.** Defines the triad in 10 patients/6 families; shows reduced TMEM94 expression, dysregulated growth/proliferation transcriptome, and embryonic-lethal KO mouse.
Fetal Anomalies Associated with Novel Pathogenic Variants in TMEM94 (Al-Hamed et al., 2020)	32825426	**Independent confirmation + prenatal expansion.** Two novel homozygous LoF variants via trio-ES + CMA; extends phenotype to the fetal period.
An investigation of the etiology and follow-up findings in 35 children with overgrowth syndromes… (Yüksel Ülker et al., 2023)	36919607	**Phenotype expansion.** TMEM94 variant in 1/35 OGID children; broadens spectrum toward overgrowth–ID.
ERMA (TMEM94) is a P-type ATPase transporter for Mg^{2+} (Vishnu et al., Mol Cell 2024)	38513662	**Mechanistic keystone.** Identifies TMEM94 as the ER Mg^{2+} P-type ATPase ERMA; links its loss to cardiomyocyte Ca^{2+} dysfunction via mouse $Erma^{+/-}$ and iPSC-CM silencing.

Limitations and Knowledge Gaps

1. **Tiny evidence base.** ~12 patients across 4 papers; frequencies and prognosis are poorly quantified, and no formal epidemiology or natural-history data exist.
2. **Mechanistic gaps for brain and face.** The ERMA/Mg^{2+} mechanism is demonstrated for the heart; the causal path to developmental delay and craniofacial dysmorphism remains inferential (via transcriptomic dysregulation).
3. **Dosage paradox unresolved.** Mouse haploinsufficiency vs human recessive inheritance is not explained.
4. **No full-recapitulation animal model.** Homozygous-null mouse is embryonic lethal; no viable model reproduces the complete human triad.

5. **No biomarkers, no targeted therapy, no trials.** Management is empirical/supportive.
6. **Localization nuance.** Early description as a "nuclear" transmembrane protein vs the functionally validated ER localization should be reconciled in annotations (ER is the correct functional site).

Proposed Follow-up Experiments / Actions

1. **Conditional/hypomorphic mouse models** (cardiac-, neural-crest-, and neuron-specific *Erma* conditional knockouts, plus hypomorphic alleles mimicking human truncations) to bypass embryonic lethality and model the full triad.
2. **Directly test ER Mg^{2+}/Ca^{2+} handling in patient-derived iPSC cardiomyocytes and neurons/organoids**, and assess whether Mg^{2+} supplementation or Mg^{2+}-channel modulation rescues Ca^{2+}-cycling defects (tests the therapeutic hypothesis).
3. **Mechanistically link ERMA loss to craniofacial development** using neural-crest and cranial-cartilage models with single-cell transcriptomics.
4. **Resolve the dosage paradox** by quantifying residual function / NMD escape of human truncating alleles and comparing mouse vs human cardiac dosage sensitivity.
5. **Establish a patient registry / GeneMatcher-driven cohort** to quantify phenotype frequencies, natural history, and genotype–phenotype correlations and to formalize diagnostic and management guidelines.
6. **Reconcile subcellular-localization annotations** (nuclear vs ER) in UniProt/GO to reflect the functionally validated ER-membrane localization.

Report compiled from 4 primary publications and public variant/constraint databases (ClinVar NM_014738.6; gnomAD). Evidence types: human clinical (cohort/case reports), model organism (mouse KO/heterozygote), in vitro (patient cells, iPSC-cardiomyocytes), and computational (population constraint).
